

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (ME, CL) Examination

November 2010

CL 202 Strength of Materials

Date: 18.05.2011, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) A beam of rectangular cross-section is 200 mm wide and 350 mm deep. If the section is subjected to a maximum shear force of 30 kN, find the maximum shear stress and sketch the shear stress distribution along the depth of the beam. 03
- (b) A timber beam 150 mm x 250 mm in cross-section is simply supported at its ends and has a span of 3.5 m. The maximum safe allowable stress in bending is 7500 kN/m². Find the maximum safe UDL which the beam can carry. 04
- Q - 2 (a) Derive the shear stress equation for rectangular cross-section and show that $q_{\max} = \frac{3}{2} q_{\text{avg}}$. 03
- (b) Derive equation of torsion with usual notations. 04
- (c) A solid circular shaft of steel is 50 mm in diameter. Find the power transmitted at 120 rpm, if the permissible shear stress is 62.5 MPa. 03
- (d) In a close coiled helical spring, the diameter of each coil is to be 10 times that of wire of the spring and the maximum shear stress is not to exceed 60 N/mm². Maximum permissible deflection under a load of 400 N is 10 cm. Taking the shear modulus as 9×10^4 N/cm², determine the number of coils, the diameter of the coil and energy stored in the coil. 04

OR

- Q - 2 (a) A triangular section of a beam has base width of 20 cm and altitude of 15 cm. The section is subjected to a shear force of 120 kN. Determine: (i) average shear stress, (ii) maximum shear stress, and (iii) shear stress at the Neutral axis. 03
- Sketch the shear stress distribution diagram over the depth of the section.
- (b) Calculate the maximum intensity of shear stress induced and angle of twist in degrees for a solid shaft of length 10 m and 100 mm diameter, transmitting 112.5 kW at 150 rpm. Take $N = 8.2 \times 10^4$ MPa. 04
- (c) The angle of twist of a 5 meter length of shaft whose diameter is 80 mm is observed to be 0.06 radian when the shaft is revolving at 4 Hz. If $G = 80$ GPa, find the power transmitted and maximum shear stress induced. 03
- (d) A shaft of 100 mm diameter transmits 112.5 kW power at 180 rpm. A flanged coupling is keyed to the shaft by means of a key 100 mm long and 30 mm wide. The coupling has 6 bolts of 20 mm diameter symmetrically arranged along the bolt circle of 300 mm diameter. 04

Calculate shearing stresses in the shaft, the key and the bolts of the coupling.

Q - 3

Attempt any Two

14

- (a) A point in a strained material is subjected to the stresses as shown in **Figure 1**. Find Normal (p_n), Tangential (p_t) and Resultant (p_r) stresses on a plane inclined at 30° with vertical as shown in Figure 3. Also find maximum Tangential stress ($p_{t \max}$). Use both Analytical and graphical method.
- (b) A point in a strained material is subjected to the stresses as shown in **Figure 2**. Find Normal (p_n), Tangential (p_t) and Resultant (p_r) stresses on a plane inclined at 40° with vertical as shown in Figure 2. Also find maximum Tangential stress ($p_{t \max}$). Use both Analytical and graphical method.
- (c) At a point in a strained material, the intensities of Normal stress across two planes right angles to each other as shown in **Figure 3**. Locate the principal planes and evaluate the principal Stresses (p_1 & p_2). Also find the planes of maximum shear stress ($p_{t \max}$) & its location. Find Normal (p_n), Tangential (p_t) stresses on a plane inclined at 60° with vertical as shown in Figure 3. Use both Analytical and graphical method.

SECTION - II

Q - 4 (a)

Derive: $\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$, with usual notations.

03

- (b) A flitched beam is made up of two timber joists 10 cm wide and 26 cm deep with a 2 cm thick steel plate 16 cm deep placed symmetrically between them and firmly attached to both. The plate is recessed into grooves cut in the inner faces of the joists so that the overall dimensions of the built-up section may be taken as 20 cm* 26 cm. Calculate the moment of resistance of the combined section when the maximum bending stress in timber is 9 MN/m^2 . What is then maximum stress in steel? Take $E_s = 20 E_w$.

04

Q - 5 (a) State the assumptions made in the theory of simple bending.

03

(b) Find the moment of inertia of section shown hatched about base for **Figure 4**.

04

(c) For the section shown in **Figure 5** find the moment of inertia about the XX axis.

03

(d) Find moment of inertia at the base of a section shown in **Figure 6**.

04

OR

Q - 5 (a) Over what length of span may a rectangular beam of section $250 \text{ mm} \times 100 \text{ mm}$ support a load of 4 kN per meter run, if the bending stress is not to exceed 12 N/mm^2 .

03

(b) Find the moment of inertia of section shown hatched about base for **Figure 7**.

04

(c) State and prove the parallel axis theorem.

03

(d) Determine the moment of inertia of the area shown in **Figure 8** about XX-axis.

04

Q - 6

Attempt any Two

14

(a) Draw Shear Force and Bending Moment Diagram for the beam shown in **Figure 9**.

(b) Draw Shear Force and Bending Moment Diagram for the beam shown in **Figure 10**. Also locate the point of maximum bending moment.

(c) Draw Shear Force and Bending Moment Diagram for the beam shown in **Figure 11**.

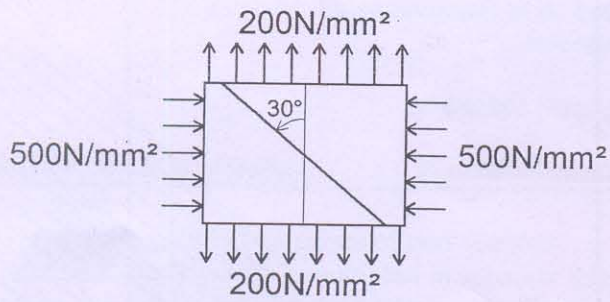


Fig. 1

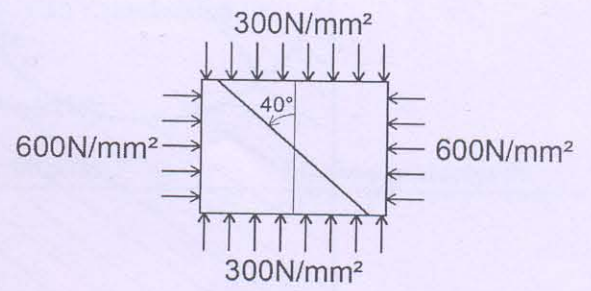


Fig. 2

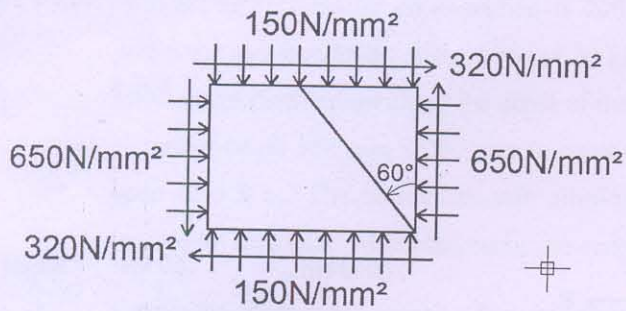


Fig. 3

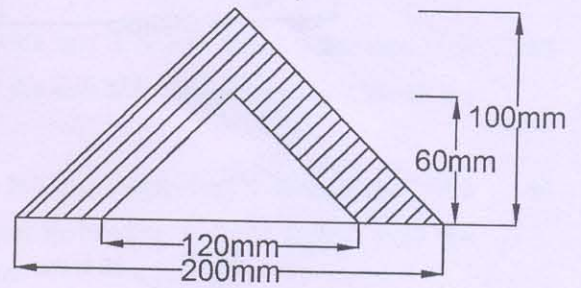


Fig. 4

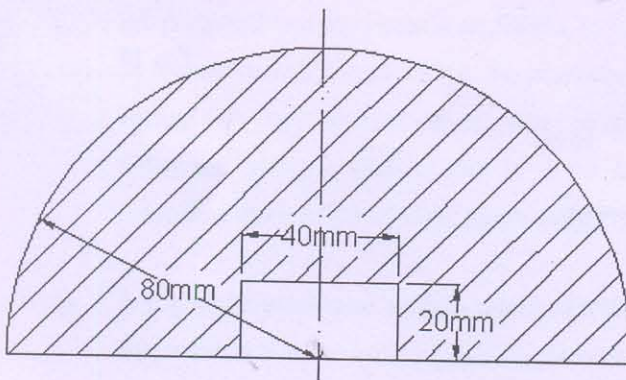


Fig. 5

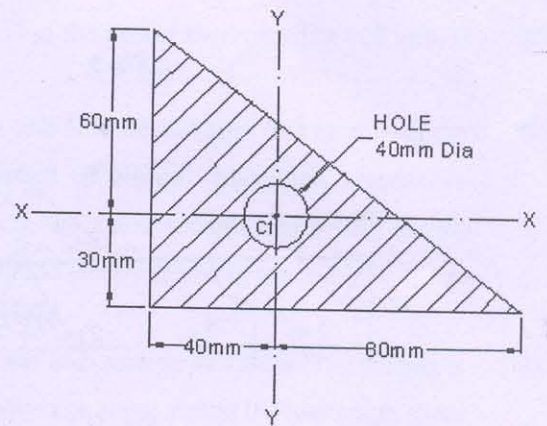


Fig. 6

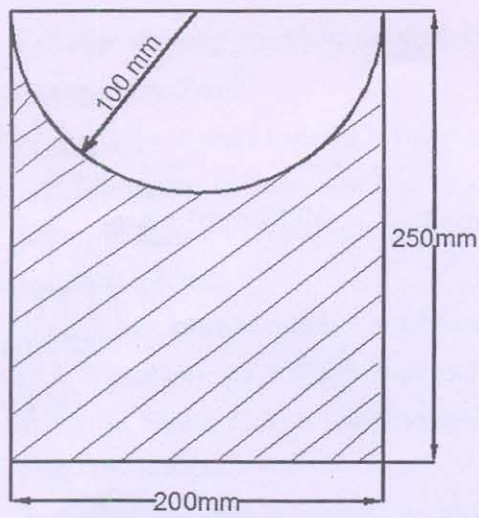


Fig. 7

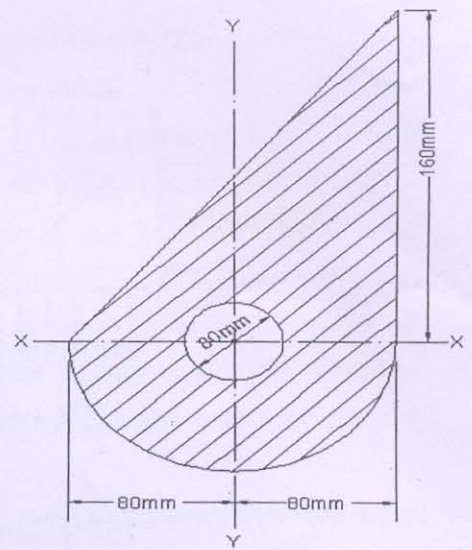


Fig. 8

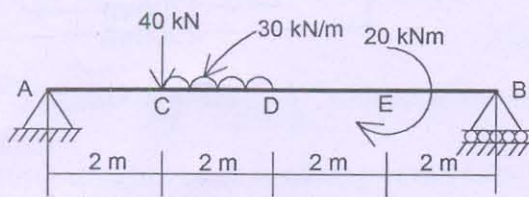


Fig. 9

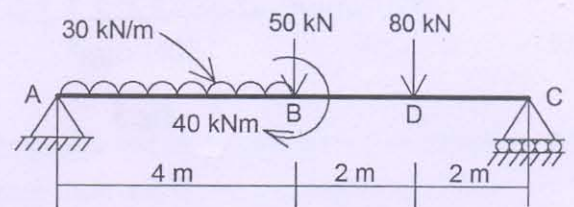


Fig. 10

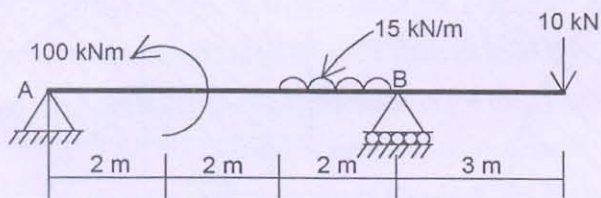


Fig. 11